DETERMINANTS · COFACTORS

The cofactor construction chain

Six definitions in strict order, each built from the one above it. Read alone they look like six pieces of terminology; read as a chain they are one construction, and the last step is what the first five were for.

6
STEPS

BUILDING A NUMBER 3

1 **Minor**

a number; unsigned

$$M_{ij} = \det(A \text{ with row } i, \text{ column } j \text{ deleted})$$

The raw building block, and note it is itself a determinant — the definition is recursive, which is where the cost comes from. For a 3×3 matrix each minor is a 2×2 determinant; for a 4×4 each is a 3×3, and so on down.

2 **Cofactor**

a number; signed

$$C_{ij} = (-1)^{i+j} M_{ij}$$

The minor with a checkerboard sign attached. The sign is not decoration — it is what makes the expansion below add up to the determinant rather than to nothing in particular, and dropping it is the most common error in the whole construction.

3 **Laplace expansion**

any single row or column

$$\det(A) = \sum_j a_{ij} C_{ij}$$

Where the cofactors do their work. Any row or column gives the same answer — a consequence of transpose invariance — so choose the one with the most zeros, since each zero entry removes an entire minor from the calculation.

BUILDING A MATRIX 3

4 **Cofactor matrix**

an $n \times n$ matrix

$$\operatorname{cof}(A)_{ij} = C_{ij}$$

Every cofactor collected in position. On its own it does nothing; it exists to be transposed in the next step, which is the only reason to compute all n^2 of them rather than the single row the expansion needs.

5 **Adjugate**

$A \cdot \operatorname{adj}(A) = \det(A) \cdot I$

$$\operatorname{adj}(A) = \operatorname{cof}(A)^{\mathsf{T}}$$

The transpose, and the identity beside it is the whole point of the chain. It holds for **every** square matrix including singular ones — when $\det(A) = 0$ it says $A \cdot \operatorname{adj}(A) = O$, which is a real statement rather than a failure.

6 **Inverse formula**

$\det(A) \neq 0$

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$$

Divide the identity above by $\det(A)$ and the inverse falls out. This is the end of the chain and its justification — an explicit formula for every entry of the inverse in terms of the entries of A.

Nothing in this chain is a good way to compute anything. A minor is a determinant, so each level multiplies the work by n — which is why row reduction computes determinants and factorization computes inverses. The chain earns its place by giving a closed form, not a method.